Sequences & Patterns

Arithmetic sequences

1 Find the 10th term of the sequence $5, 8, 11, 14, \ldots$

2 Find the 15th term of the sequence $7, 12, 17, 22, \ldots$

3 An arithmetic sequence has first term 4 and common difference 6. Find the 20th term.

4 The 5th term of an arithmetic sequence is 23, and the first term is 3. Find the common difference.

5 An arithmetic sequence has first term 7 and common difference 5. Which term equals 82?

6 The 8th term of an arithmetic sequence is 34, and the common difference is 4. Find the first term.

Geometric sequences

7 Find the next term of the sequence $2, 4, 8, 16, \ldots$

8 Find the 6th term of the sequence $3, 6, 12, 24, \ldots$

9 A geometric sequence has first term 5 and common ratio 3. Find the 5th term.

10 The 4th term of a geometric sequence is 54, and the first term is 2. Find the common ratio.

11 A geometric sequence has 2nd term 6 and 3rd term 18. Find the first term and the common ratio.

Identifying the type of sequence

12 Is $5, 10, 20, 40, \ldots$ arithmetic, geometric, or neither?

13 Is $6, 10, 14, 18, \ldots$ arithmetic, geometric, or neither?

14 Is $1, 4, 9, 16, \ldots$ arithmetic, geometric, or neither?

15 Is $100, 90, 81, 72.9, \ldots$ arithmetic, geometric, or neither?

Finding missing terms

16 Find the missing term: $3,\ \_\_,\ 11,\ 15$ (arithmetic).

17 Find the missing term: $4,\ \_\_,\ 36$ (geometric, all terms positive).

18 Insert two arithmetic means between 5 and 20, forming the four-term sequence $5, \_\_, \_\_, 20$.

19 The sequence $x,\ 2x+1,\ 4x-1$ is arithmetic. Find x.

20 Find the two missing terms of the geometric sequence $2,\ \_\_,\ \_\_,\ 54$.

Sum of an arithmetic series

21 Find the sum of the first 10 terms of the sequence $2, 5, 8, 11, \ldots$

22 Find the sum of the first 20 positive integers.

23 Find the sum of the first 12 terms of the sequence $10, 7, 4, 1, \ldots$

24 An arithmetic sequence has first term 3 and common difference 4. Find the sum of its first 15 terms.

25 Find the sum of the first 25 positive even integers ($2, 4, \ldots, 50$).

Special sequences

26 The sequence $1, 4, 9, 16, 25, \ldots$ consists of perfect squares. Find the 8th term.

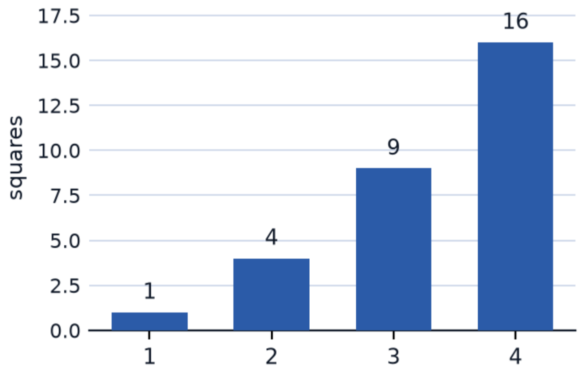
27 Find the 6th triangular number, from the sequence $1, 3, 6, 10, 15, \ldots$

28 A sequence starts $2, 3, 5, 8, 13, \ldots$, where each term is the sum of the two terms before it. Find the next two terms.

29 Find the 7th term of the sequence of powers of 2: $1, 2, 4, 8, 16, \ldots$

Visual and growing patterns

30 The bar chart shows the number of small squares used to build patterns 1 through 4. Predict the number of squares in pattern 6, and state the rule.



31 A pattern of dots grows as $3, 7, 11, 15, \ldots$ (each stage adds 4 dots). Find the number of dots in stage 10.

32 A row of matchstick triangles uses $3, 5, 7, 9, \ldots$ matchsticks for $1, 2, 3, 4$ triangles. Find the number of matchsticks for 15 triangles in a row.

33 A row of matchstick hexagons uses $6, 11, 16, 21, \ldots$ matchsticks for $1, 2, 3, 4$ hexagons. Find the matchsticks needed for 9 hexagons in a row.

Word problems

34 A theater has 20 seats in the first row, and each following row has 3 more seats than the row before it. How many seats are in the 12th row?

35 A ball is dropped and after each bounce rises to half its previous height. If it first rises to 80 m after being dropped, how high does it rise after the 4th bounce?

36 A savings plan deposits 100 riyals in week 1, then increases the deposit by 20 riyals each following week. Find the total saved after 8 weeks.

37 A bacteria culture starts with 50 bacteria and doubles every hour. How many bacteria are there after 6 hours?

More on series and sequences

38 Find the sum of the first 8 terms of the geometric sequence $1, 2, 4, 8, \ldots$

39 A geometric sequence has first term 4 and common ratio 2. Find the sum of its first 6 terms.

40 An arithmetic sequence has 3rd term 11 and 7th term 27. Find the common difference and the first term.

41 The nth term of a sequence is $T_n = 3n - 2$. Find the 12th term, and state whether the sequence is arithmetic.

Pattern reasoning

42 A sequence of tile patterns is built so that stage n uses $n^2 + 1$ tiles. How many tiles are needed for stage 7?

43 The number of diagonals in a polygon with n sides is $\frac{n(n-3)}{2}$. Find the number of diagonals in an octagon ($n = 8$).

44 A sequence follows the rule "double and add 1", starting at 3 as the 1st term. Find the 4th term.

45 The number of handshakes when n people each shake hands with every other person once is $\frac{n(n-1)}{2}$. Find the number of handshakes among 10 people.

MCQ — Arithmetic sequences

46 Find the 12th term of the sequence $4, 9, 14, 19, \ldots$

a. 49

b. 54

c. 64

d. 59

47 Find the 18th term of the sequence $6, 13, 20, 27, \ldots$

a. 120

b. 132

c. 125

d. 118

48 An arithmetic sequence has first term 9 and common difference 4. Find the 25th term.

a. 100

b. 105

c. 109

d. 96

49 The 6th term of an arithmetic sequence is 41, and the first term is 6. Find the common difference.

a. 7

b. 8

c. 6

d. 35

50 An arithmetic sequence has first term 10 and common difference 6. Which term equals 100?

a. 17th

b. 90th

c. 15th

d. 16th

51 The 10th term of an arithmetic sequence is 52, and the common difference is 5. Find the first term.

a. 12

b. 2

c. 7

d. 45

MCQ — Geometric sequences

52 Find the next term of the sequence $3, 6, 12, 24, \ldots$

a. 30

b. 27

c. 36

d. 48

53 Find the 7th term of the sequence $2, 6, 18, 54, \ldots$

a. 486

b. $4{,}374$

c. $1{,}458$

d. 729

54 A geometric sequence has first term 4 and common ratio 2. Find the 6th term.

a. 256

b. 32

c. 128

d. 64

55 The 5th term of a geometric sequence is 162, and the first term is 2. Find the common ratio.

a. 3

b. 9

c. 4

d. 2

56 A geometric sequence has 3rd term 20 and 4th term 40. Find the first term and the common ratio.

a. $a_1 = 5, r = 2$

b. $a_1 = 20, r = 2$

c. $a_1 = 5, r = 4$

d. $a_1 = 10, r = 2$

57 A geometric sequence has first term 7 and common ratio 2. Find the 8th term.

a. 448

b. $1{,}792$

c. 112

d. 896

MCQ — Identifying the type of sequence

58 Is $7, 14, 28, 56, \ldots$ arithmetic, geometric, or neither?

a. Neither

b. Arithmetic

c. Geometric

d. Both arithmetic and geometric

59 Is $8, 13, 18, 23, \ldots$ arithmetic, geometric, or neither?

a. Both arithmetic and geometric

b. Geometric

c. Arithmetic

d. Neither

60 Is $2, 5, 10, 17, \ldots$ arithmetic, geometric, or neither?

a. Geometric

b. Arithmetic

c. Both arithmetic and geometric

d. Neither

61 Is $200, 180, 162, 145.8, \ldots$ arithmetic, geometric, or neither?

a. Both arithmetic and geometric

b. Arithmetic

c. Geometric

d. Neither

MCQ — Finding missing terms

62 Find the missing term: $6,\ \_\_,\ 16,\ 21$ (arithmetic).

a. 9

b. 13

c. 11

d. 10

63 Find the missing term: $5,\ \_\_,\ 80$ (geometric, all terms positive).

a. 42.5

b. 20

c. 16

d. 40

64 Insert two arithmetic means between 8 and 29, forming the four-term sequence $8, \_\_, \_\_, 29$.

a. $15, 22$

b. $14, 21$

c. $12.5, 20.5$

d. $15, 21$

65 The sequence $2x,\ 3x+4,\ 5x+2$ is arithmetic. Find x.

a. 4

b. 2

c. 8

d. 6

66 Find the two missing terms of the geometric sequence $3,\ \_\_,\ \_\_,\ 81$.

a. $12, 27$

b. $9, 30$

c. $27, 9$

d. $9, 27$

67 Find the missing term: $\_\_,\ 12,\ 19,\ 26$ (arithmetic).

a. 7

b. 9

c. 4

d. 5

MCQ — Sum of an arithmetic series

68 Find the sum of the first 12 terms of the sequence $3, 7, 11, 15, \ldots$

a. 312

b. 300

c. 264

d. 282

69 Find the sum of the first 30 positive integers.

a. 900

b. 450

c. 435

d. 465

70 Find the sum of the first 15 terms of the sequence $12, 9, 6, 3, \ldots$

a. -135

b. -153

c. -108

d. 135

71 An arithmetic sequence has first term 5 and common difference 3. Find the sum of its first 20 terms.

a. 685

b. 670

c. 640

d. 625

72 Find the sum of the first 30 positive even integers $(2, 4, \ldots, 60)$.

a. 465

b. 960

c. 900

d. 930

73 Find the sum of the first 10 terms of the sequence $20, 15, 10, 5, \ldots$

a. -25

b. -50

c. 25

d. 5

MCQ — Sum of a geometric series

74 Find the sum of the first 6 terms of the sequence $2, 4, 8, 16, \ldots$

a. 128

b. 124

c. 126

d. 62

75 A geometric sequence has first term 5 and common ratio 3. Find the sum of its first 5 terms.

a. 605.5

b. $1{,}210$

c. 363

d. 605

76 Find the sum of the first 4 terms of the sequence $1, 3, 9, 27, \ldots$

a. 40

b. 81

c. 27

d. 39

77 A geometric sequence has first term 6 and common ratio 2. Find the sum of its first 7 terms.

a. 768

b. 762

c. 384

d. 381

MCQ — Special sequences

78 The sequence $1, 4, 9, 16, 25, \ldots$ consists of perfect squares. Find the 10th term.

a. 100

b. 121

c. 90

d. 81

79 Find the 7th triangular number, from the sequence $1, 3, 6, 10, 15, 21, \ldots$

a. 28

b. 25

c. 36

d. 21

80 A sequence starts $3, 5, 8, 13, 21, \ldots$, where each term is the sum of the two terms before it. Find the 8th term.

a. 55

b. 89

c. 144

d. 76

81 Find the 8th term of the sequence of powers of 3: $1, 3, 9, 27, 81, \ldots$

a. 729

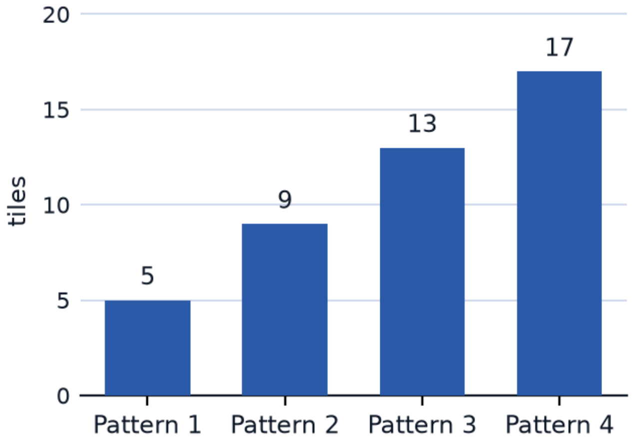
b. $6{,}561$

c. $2{,}184$

d. $2{,}187$

MCQ — Visual and growing patterns

82 The bar chart shows the number of tiles used to build patterns 1 through 4. Predict the number of tiles in pattern 8.



a. 37

b. 40

c. 29

d. 33

83 A pattern of dots grows as $2, 6, 10, 14, \ldots$ (each stage adds 4 dots). Find the number of dots in stage 10.

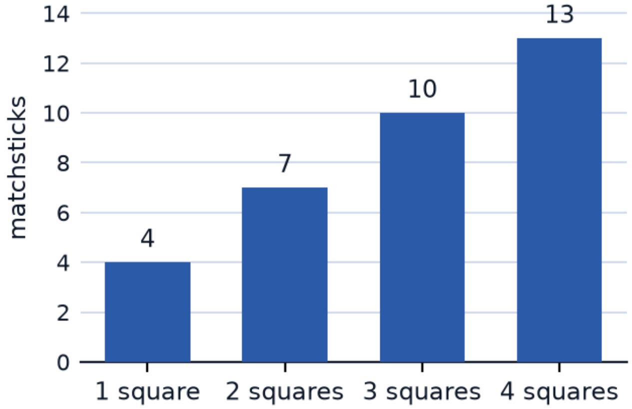
a. 38

b. 40

c. 34

d. 42

84 The bar chart shows the number of matchsticks used to build squares in a row, for 1 through 4 squares. Find the number needed for 12 squares.



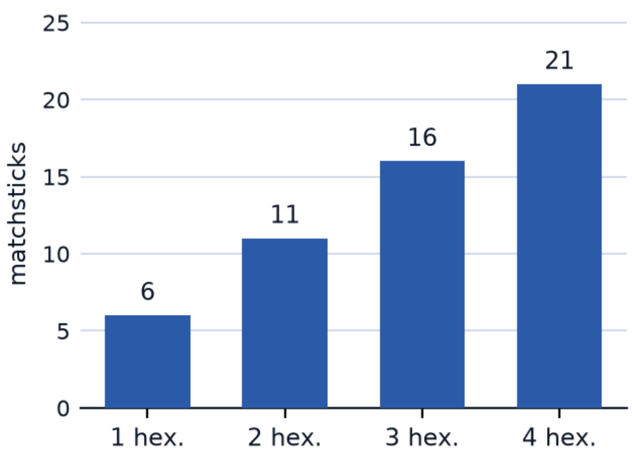
a. 34

b. 43

c. 40

d. 37

85 The bar chart shows the number of matchsticks used to build hexagons in a row, for 1 through 4 hexagons. Find the number needed for 9 hexagons.



a. 41

b. 51

c. 50

d. 46

MCQ — Word problems

86 A theater has 18 seats in the first row, and each following row has 4 more seats than the row before. How many seats are in row 15?

- **a.** 78
- **b.** 70
- **c.** 56
- **d.** 74

87 A ball is dropped and after each bounce rises to half its previous height. If it first rises to 120 cm after the first bounce, how high does it rise after the 5th bounce?

- **a.** 60
- **b.** 15
- **c.** 3.75
- **d.** 7.5

88 A savings plan deposits 80 riyals in week 1, then increases the deposit by 15 riyals each following week. Find the deposit in week 10.

- **a.** 215
- **b.** 200
- **c.** 230
- **d.** 935

89 A bacteria culture starts with 40 bacteria and triples every hour. How many bacteria are there after 4 hours?

a. 1,080

b. 3,240

c. 9,720

d. 480

MCQ — Pattern reasoning

90 A sequence of tile patterns is built so that stage n uses $2n^2 + 1$ tiles. How many tiles are needed for stage 5?

a. 50

b. 26

c. 51

d. 41

91 The number of diagonals in a polygon with n sides is $\frac{n(n-3)}{2}$. Find the number of diagonals in a polygon with 9 sides.

a. 36

b. 27

c. 21

d. 54

92 A sequence follows the rule \"triple and subtract 2\", starting at 4 as the 1st term. Find the 4th term.

a. 34

b. 28

c. 82

d. 250

93 The number of handshakes when n people each shake hands with every other person once is $\frac{n(n-1)}{2}$. Find the number of handshakes among 12 people.

a. 66

b. 132

c. 56

d. 78

94 The nth term of a sequence is $T_n = 4n - 3$. Find the 15th term, and state whether the sequence is arithmetic or geometric.

a. 57, geometric

b. 53, arithmetic

c. 60, arithmetic

d. 57, arithmetic

95 The nth term of a sequence is $T_n = 5(2)^{n-1}$. Find the 6th term, and state whether the sequence is arithmetic or geometric.

a. 160, geometric

b. 320, geometric

c. 160, arithmetic

d. 80, geometric